Sessional Test-I

Dated: 09-09-2024, Full Marks: 30

Course Name: Physics, Course Code: BPY23111T

Programme: B. Tech In Civil Engineering

1. Fill in the blanks: (1x5=5)
   (i) $\hat{\imath}.\hat{\imath}$ = ____ (ii) $\hat{k}$ x $\hat{k}$ = __ (iii) $\vec{\nabla}.\vec{\nabla}$ = ____ (iv) $\hat{k}$ x $\hat{\imath}$ (v) $\hat{\imath}.\hat{\jmath}$= ____
2. Answer the following questions briefly (2x5=10)
   (i) What do you mean by divergence of a vector field?
   (ii) State Gauss' law on divergence.
   (iii) Write the Poisson's and Laplace equations.
   (iv) Find the divergence of the vector, $\vec{V}=2x^2y\,\hat{\imath}- 4xz\,\hat{\jmath} + 9xyz^3\,\hat{k}$.
   (v) Find the gradient of a scalar function, $A=3xy+2zx+x^2z$
3. Answer any **one** from the following: (5)
   (i) State Ampere's law and explain its inconsistency.
   (ii) Write the Maxwells's equations and explain in brief about each vector involved in them.
4. Answer any **one** of the following (10)
   (i) Explain Maxwell's modification of Ampere's law.
   (ii) Explain the physical significance of Maxwell's equations.